

Notes: Huck (RFS, 2024)

The Psychological Externalities of Investing: Evidence from Stock Returns and Crime

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1 Big picture

- **Question.** Do daily stock market returns affect psychological well-being, and do those effects extend beyond stockholders to people who do not own stocks?
- **Measurement idea.** The paper uses crime as a revealed-behavior proxy for psychological well-being. If a market movement changes mood, stress, relative status, or perceived economic position, one possible behavioral margin is an immediate change in violent crime.
- **Empirical setting.** The paper studies over 55 million reported daily crime incidents from the FBI's National Incident-Based Reporting System (NIBRS), covering roughly 2,700 law-enforcement agencies from 1991 to 2015. The core outcome is the daily crime rate per 100 million people at the agency level.
- **Investor versus noninvestor split.** Locations are sorted by median personal income. High-income locations are interpreted as more likely to contain investors; low-income locations are interpreted as more likely to contain noninvestors. The paper also validates this split using alternative stockholder-status proxies based on dividend income and Survey of Consumer Finances stockholding propensities.
- **Main result.** A one-standard-deviation increase in the daily market return is associated with a **decrease** in violent crime in high-income locations and an **increase** in violent crime in low-income locations. In the baseline violent-crime specification, the estimates are about -12.5 violent crimes per 100 million people in high-income locations and $+15.2$ in low-income locations.
- **Economic interpretation.** For investors, positive stock returns raise wealth and appear to improve well-being. For noninvestors, positive stock returns may signal that others are getting richer while they are not, worsening relative status and well-being.
- **Timing result.** The effect is contemporaneous. Daily returns are related to same-day violent crime, not to violent crime on surrounding days. Within the day, the high-income response appears soon after the market close, while the low-income response appears later in the evening.
- **Identification reinforcement.** Local earnings surprises produce a similar pattern when the paper includes date fixed effects. This helps separate the mechanism from common daily shocks such as macroeconomic announcements, war, or national news.
- **Mechanism checks.** The noninvestor result is not explained by alcohol-suspected violent crime, by local alcohol/drug usage controls, or by more public targets after good market news. The pattern survives when focusing on at-home violence by known offenders.
- **Takeaway.** Stock market fluctuations have psychological externalities. Even people who do not participate in the stock market may be affected by stock-market news through social comparison and relative wealth.

2 Data and measurement (key objects)

2.1 Crime data

- **Source.** Daily crime incidents come from NIBRS, a voluntary FBI database that records incident-level reports by local law-enforcement agencies.

- **Sample period.** 1991–2015.
- **Raw and filtered scale.** NIBRS contains more than 80 million incidents; after reporting-consistency filters, the study uses more than 55 million incidents, or more than 39 million incidents on trading days.
- **Unit of observation.** Police agency/day. Agencies are city police departments or county sheriff’s offices.
- **Crime-day convention.** Crime day t begins at 6:00 AM local time on calendar day t and ends at 5:59 AM on day $t + 1$. This aligns crime timing with the market day and includes the evening after the close.
- **Outcome variable.** The main crime rate is

$$y_{i,t} = 100,000,000 \times \frac{\text{incidents}_{i,t}}{\text{population}_i},$$

where i indexes the agency and t indexes the trading day.

- **All crimes.** The broad measure aggregates all 46 NIBRS Group A offenses, including crimes against persons, property, and society.
- **Violent crime.** The main analysis focuses on assaults and homicides. This is important because violent crime is more likely to occur close to home and involve known offenders, reducing the concern that crimes in high-income areas are actually committed by low-income outsiders.

Interpretation. Crime is not a direct survey measure of happiness. It is a costly revealed action. That makes it noisy but also valuable: it is high-frequency, standardized across locations, and less vulnerable to survey framing or recall bias.

2.2 Investor and noninvestor status

- The paper uses 2012 American Community Survey microdata to estimate median personal income for each city/county.
- **High-income agencies** are the top tercile of agency locations by income. These are treated as more likely investor locations.
- **Low-income agencies** are the bottom tercile. These are treated as more likely noninvestor locations.
- The income split is not the only measure. The paper also uses dividend income from IRS data and estimated stockholder propensities from SCF/ACS demographic mappings. The main pattern is similar under these alternatives.

Interpretation. The investor split is location-based, not person-level. This is a key limitation, but violent crime and the alternative stockholder-status proxies help reduce measurement concerns.

2.3 Financial variables

- **Market returns.** Daily U.S. stock market returns come from Kenneth French’s data library.

- **Standardization.** Returns are divided by their trailing 252-day standard deviation:

$$r_t = \frac{R_t}{\hat{\sigma}_{t-252:t-1}}.$$

Therefore, a one-unit change in r_t is roughly a one-standard-deviation daily market return.

- **Local earnings surprises.** To test whether salient firm-specific news generates similar psychological responses, the paper constructs regional standardized unexpected earnings:

$$\text{SUE}_{j,q} = \frac{\text{Earnings}_{j,q} - \text{Earnings}_{j,q-4}}{\text{Price}_{j,q}},$$

then averages across salient firms headquartered in the region.

- **Announcement returns.** The paper also uses the two-day cumulative return around earnings announcements as an alternative local-news measure.

3 Empirical specifications (all equations + interpretation)

3.1 Main daily crime-return regression

The baseline regression is

$$y_{i,t} = \beta r_t + \Gamma X_{i,t} + \theta_{i,a(t)} + \mu_{i,m(t)} + \omega_{i,w(t)} + \delta_{i,d(t)} + T_t + \varepsilon_{i,t}. \quad (1)$$

- $y_{i,t}$ is the crime rate in agency i on day t .
- r_t is the standardized daily market return.
- $X_{i,t}$ includes controls for weather, pollution, moon phases, geomagnetic storms, Seasonal Affective Disorder, sunset hour, daylight-saving changes, and local NFL/MLB game outcomes.
- $\theta_{i,a(t)}$ are agency-by-year fixed effects.
- $\mu_{i,m(t)}$ are agency-by-month-of-year fixed effects.
- $\omega_{i,w(t)}$ are agency-by-week-of-month fixed effects.
- $\delta_{i,d(t)}$ are agency-by-day-of-week fixed effects.
- T_t includes turn-of-month and holiday effects.
- Standard errors are clustered by agency and date.

Read the coefficient. β measures the change in daily crime per 100 million people associated with a one-standard-deviation market return, within the same agency and conditional on rich seasonality, local time patterns, and controls.

Why this specification is powerful. The effect is identified from high-frequency movements in the market, not from long-run differences across cities. Agency-by-year fixed effects absorb persistent and annually varying local factors such as local economic conditions, demographics, policing intensity, and reporting culture.

3.2 Investor versus noninvestor heterogeneity

The main regression is estimated separately for high-, medium-, and low-income agencies. The comparison is:

$$\beta^{High} \quad \text{versus} \quad \beta^{Low}.$$

Interpretation. If the stock market affects only investors through wealth, then we should expect a response mainly in high-income locations. If noninvestors are also affected through relative wealth or social comparison, low-income locations should respond as well, possibly with the opposite sign.

3.3 Local earnings-surprise regression

The local-news test is

$$y_{i,t} = \beta \text{SUE}_{i,t} + \Gamma X_{i,t} + \eta_i + \tau_t + \varepsilon_{i,t}. \quad (2)$$

- $\text{SUE}_{i,t}$ is the regional average standardized unexpected earnings for firms headquartered near agency i .
- η_i are agency fixed effects.
- τ_t are date fixed effects.

Read the coefficient. β measures whether local salient firm news moves local violent crime relative to other locations on the same day.

Why date fixed effects matter. Date fixed effects absorb marketwide daily shocks, macro news, national news, and any common event affecting all locations. The remaining variation comes from local earnings news.

3.4 Relative-wealth interpretation

The paper motivates the empirical pattern with a crime-propensity equation:

$$p = p_0 + \lambda u' \left(\frac{w}{\bar{w}} \right), \quad \lambda > 0, \quad (3)$$

where p is the probability or propensity of crime, p_0 is a baseline propensity, w is own wealth, and $\bar{w}$ is reference-group or average wealth.

Interpretation.

- Investors benefit from positive market returns, so $w/\bar{w}$ rises or at least does not fall as much. Marginal utility falls, and violent crime declines.
- Noninvestors do not receive direct stock-market wealth gains. If market gains raise the wealth of others, noninvestors may feel relatively worse off. Marginal utility rises, and violent crime increases.

4 Identification and mechanisms (what makes the estimates credible)

4.1 High-frequency timing

- The paper uses daily returns and daily crime. Slow-moving confounders like recessions, local income trends, policing budgets, or demographic change are unlikely to move at daily frequency.
- Leads and lags show that the relationship is mostly same-day. Past and future returns do not generally explain current violent crime.
- Hourly crime data show that the response is concentrated after the market close. This supports a psychological-information channel rather than a generic long-run economic channel.

4.2 Extensive controls and fixed effects

- The fixed effects absorb local seasonality at several levels: year, month-of-year, week-of-month, and day-of-week by agency.
- Controls address factors known to affect mood, crime, or returns: weather, pollution, moon phase, geomagnetic storms, SAD, daylight saving time, and sports outcomes.
- Two-way clustering handles serial correlation within agencies and cross-sectional correlation across agencies on the same day.

4.3 Why focus on violent crime

- Property crimes in rich areas could be committed by poorer outsiders, creating mismeasurement of offender income and investor status.
- Violent crimes, especially assaults and homicides, are more likely to occur at or near home and involve known offenders.
- The paper further narrows the outcome to at-home violence by known offenders and finds the same high-versus-low income pattern.

4.4 Local earnings surprises as a common-shock test

- A possible alternative is that market returns and crime are both responding to unobserved national news.
- The earnings-surprise design uses date fixed effects, so the comparison is across regions on the same day.
- The result has the same sign pattern: good local firm news lowers violent crime in high-income locations and raises it in low-income locations.

4.5 Limitations

- Crime is a proxy for psychological well-being, not a direct measure of utility.

- Investor status is inferred from location income, not directly observed at the offender level.
- NIBRS participation is voluntary, and reporting behavior could vary over time. The paper’s fixed effects and focus on serious violent crimes reduce but do not eliminate this concern.
- The relative-wealth mechanism is strongly suggestive but not uniquely identified. Positive market returns can also signal macroeconomic prospects, news salience, and other psychological channels.

5 Main empirical results

5.1 All crimes

- Using all 46 NIBRS Group A offenses, a one-standard-deviation market return is associated with an increase of about 23.5 crimes per 100 million people, or roughly 10 basis points relative to the mean crime rate.
- The all-crime effect is stronger in low-income locations. This is consistent with the idea that many crimes are committed by noninvestors and that positive market news can worsen relative status for them.
- The all-crime result is less clean than violent crime because property crimes may occur in locations that differ from the offender’s location or status.

5.2 Violent crime

- For high-income locations, a one-standard-deviation market return is associated with a decrease of about 12.5 violent crimes per 100 million people, equal to about -37 basis points relative to the high-income mean.
- For low-income locations, the same market return is associated with an increase of about 15.2 violent crimes per 100 million people, equal to about $+25$ basis points relative to the low-income mean.
- The difference between high- and low-income coefficients is statistically significant.
- When stockholder status is measured using the intersection of income, dividend income, and estimated stockholding propensity, the contrast becomes even sharper: about -16.3 in high-status locations and $+25.9$ in low-status locations.

Interpretation. The same market movement appears to have opposite psychological effects depending on whether the local population is likely to benefit directly from the stock market.

5.3 Timing

- Leads and lags show that market returns on day t predict violent crime on day t , but not on adjacent days.
- For high-income locations, the after-close effect is strongest between 4 PM and 7 PM.
- For low-income locations, the effect appears later in the evening and into the night, consistent with slower diffusion of stock-market news or delayed response.

- Randomization tests assign market returns to random trading dates. The placebo coefficients are centered near zero, and the actual coefficients lie in the tails of the placebo distributions.

5.4 Earnings surprises

- Local standardized unexpected earnings produce the same cross-income pattern as market returns.
- In high-income locations, a one-standard-deviation increase in SUE corresponds to about -19.1 violent crimes per 100 million people.
- In low-income locations, it corresponds to about $+25.7$ violent crimes per 100 million people.
- Two-day announcement returns produce a similar result: negative in high-income locations and positive in low-income locations.

Interpretation. Salient local firm news also changes local behavior. This supports the idea that financial information itself, not only broad macroeconomic news, affects well-being.

5.5 Alternative explanations

- **Alcohol channel.** Market returns are not significantly related to alcohol-suspected violent crime. Adding local alcohol and drug-use controls leaves the main estimates essentially unchanged.
- **More targets in public.** The pattern survives when restricting to violent crimes committed at home by known offenders. Therefore, the low-income effect is unlikely to be driven only by more people going out after good market news.
- **Household conflict.** In high-income locations, the market-return effect is particularly strong for at-home male-on-female intimate-partner violence. This suggests that financial stress can become household conflict.

6 Figures and tables

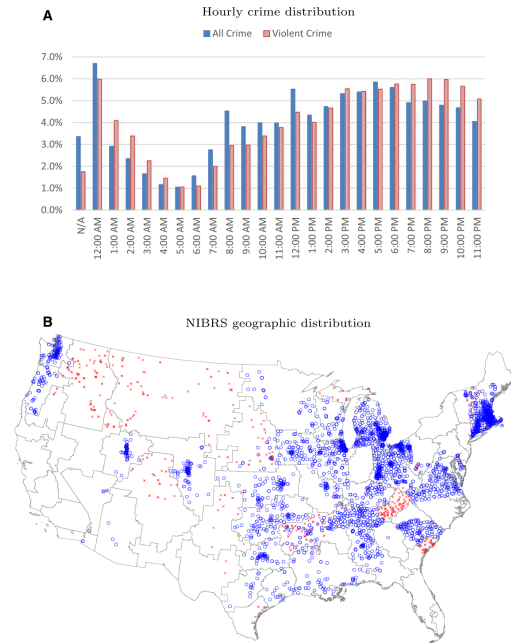
6.1 Table 1: Group A crimes in NIBRS and Figure 1: Hourly and geographic crime distribution

Table 1: Group A crimes in NIBRS

Crimes against persons	Crimes against property
Violent crime	Larceny/Theft offenses
Homicide offenses	Pocket-picking
Murder/Nonnegligent manslaughter	Purse-snatching
Negligent manslaughter	Shoplifting
Justifiable homicide	Theft from building
Assault offenses	Theft from coin-operated machine or device
Aggravated assault	Theft from motor vehicle
Simple assault	Theft of motor vehicle parts/accessories
Intimidation	All other larceny
Sex offenses, forcible	Fraud offenses
Forcible rape	False pretenses/swindle/confidence game
Forcible sodomy	Credit card/automatic teller machine fraud
Sexual assault with an object	Impersonation
Forcible fondling	Welfare fraud
Sex offenses, nonforcible	Wire fraud
Incest	Robbery
Statutory rape	Arson
Kidnapping/Abduction	Extortion/Blackmail
	Burglary/Breaking and entering
Crimes against society	Motor vehicle theft
Drug/Narcotic offenses	Counterfeiting/Forgery
Drug/Narcotic violations	Embezzlement
Drug equipment violations	Stolen property offenses
Gambling offenses	Destruction/Damage/Vandalism of property
Betting/Wagering	Bribery
Operating/Promoting gambling	
Gambling equipment violations	
Sports tampering	
Prostitution offenses	
Prostitution	
Assisting or promoting prostitution	
Pornography/Obscene material	
Weapon law violations	

List of all 46 Group A crime offenses recorded in NIBRS. Offenses are divided into three categories by NIBRS: crimes against persons, crimes against property, or crimes against society. Crimes against persons are those whose victims are always individuals. Crimes against property are those whose objective is to obtain money, property, or some other benefit. Crimes against society represent society's prohibition against engaging in certain types of activity.

Figure 1: Hourly and geographic crime distribution



Read. The crime classification table shows the scope of the all-crime outcome. The first distribution figure explains why the paper defines the crime day from 6 AM to 5:59 AM and shows the geographic spread of agencies. The key measurement choice is to separate all crimes from violent crimes because violent crime is a cleaner proxy for the offender's own psychological state.

6.2 Figure 2: NIBRS coverage and Table 2: Summary statistics

Figure 2: NIBRS coverage

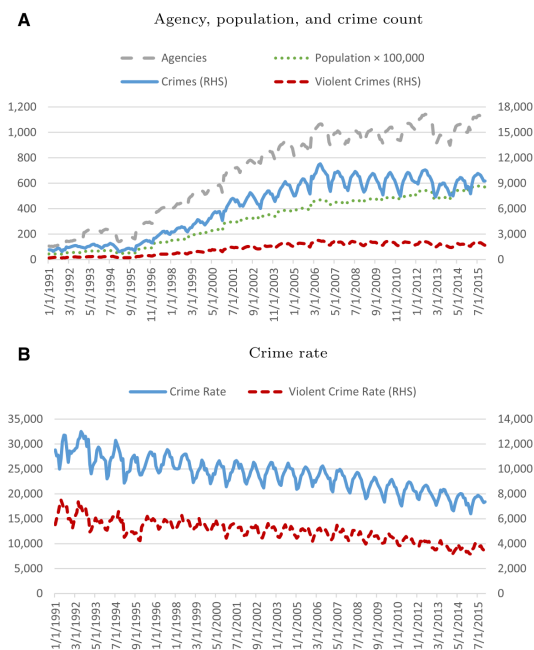


Table 2: Summary statistics

	Average	Median	SD
Crime rate	22,659	17,675	18,565
High-income	18,347	14,547	14,577
Medium-income	24,215	19,618	18,894
Low-income	27,043	21,527	21,784
Violent crime rate	4,676	2,437	7,140
High-income	3,370	1,524	5,455
Medium-income	5,077	3,021	7,199
Low-income	6,093	3,637	8,748
Market return	0.038	0.082	1.044
SUE	-0.074	0.115	4.260
Announcement ret. (0,1)	0.202	0.206	6.684
Alcohol rate	308	0	1,966
Drug rate	1,029	0	3,526

This table presents summary statistics for daily data from 1991 to 2015. High-income is defined as the top tercile of income for all agency locations, while low-income is defined as the bottom tercile. Market returns are divided by its trailing 252-day standard deviation. Crime rates are measured as the number of incidents per 100 million people. Crime rate includes all crime offenses in Table 1, while violent crime rate only includes assaults and homicides. SUE is the difference between reported quarterly earnings and earnings from the same quarter in the previous year, scaled by the quarter end price. Announcement day return is the 2-day cumulative return on announcement date t and $t+1$. Alcohol and drug rates are estimated using all alcohol and drug use suspected crime for Group A incidents, excluding violent crimes, but including the Group B crimes of driving under the influence and drunkenness. Summary statistics for control variables can be found in Table IA.1 of the Internet Appendix.

Read. Coverage grows substantially over time, from a small early sample to a much broader national panel. The summary statistics show that crime rates are higher in low-income locations, and violent crime accounts for roughly one-fifth of the all-crime rate. This matters because the main estimates are interpreted as changes relative to different baseline crime rates.

6.3 Table 3: Market returns and crime rates by income and stockholder status and Table 4: Returns: Daily lagging and leading market returns and violent crime rates

Table 3: Market returns and crime rates by income and stockholder status

A. All incidents by income and combined stockholder status

	ALL	Income			Combined SH status		
		HIGH	MED	LOW	HIGH	MED	LOW
Market return	23.53** (2.180)	0.290 (0.0301)	22.90 (1.577)	57.63*** (3.138)	-0.520 (-0.0399)	31.64 (1.377)	75.41*** (2.765)
Observations	4,344,657	1,707,967	1,482,632	1,154,058	719,463	442,420	473,592
R^2	.536	.504	.521	.524	.505	.542	.495
Controls	YES	YES	YES	YES	YES	YES	YES
Agency × Yr FE	YES	YES	YES	YES	YES	YES	YES
Agency × Mth of yr FE	YES	YES	YES	YES	YES	YES	YES
Agency × Wk of mth FE	YES	YES	YES	YES	YES	YES	YES
Agency × Day of wk FE	YES	YES	YES	YES	YES	YES	YES
Turn of mth FE	YES	YES	YES	YES	YES	YES	YES
Holiday FE	YES	YES	YES	YES	YES	YES	YES
% of avg	0.10	0.00	0.09	0.21	-0.00	0.13	0.27
$t(\text{HIGH}=\text{LOW})$		2.765			2.512		

B. Violent crime incidents by income and combined stockholder status

	ALL	Income			Combined SH status		
		HIGH	MED	LOW	HIGH	MED	LOW
Market return	-1.666 (-0.463)	-12.47*** (-3.291)	-2.816 (-0.499)	15.22** (2.043)	-16.25*** (-2.841)	-3.287 (-0.278)	25.89** (2.111)
Observations	4,344,529	1,707,967	1,482,504	1,154,058	719,463	442,420	473,592
R^2	.294	.258	.284	.279	.252	.280	.273
Controls	YES	YES	YES	YES	YES	YES	YES
Agency × Yr FE	YES	YES	YES	YES	YES	YES	YES
Agency × Mth of yr FE	YES	YES	YES	YES	YES	YES	YES
Agency × Wk of mth FE	YES	YES	YES	YES	YES	YES	YES
Agency × Day of wk FE	YES	YES	YES	YES	YES	YES	YES
Turn of mth FE	YES	YES	YES	YES	YES	YES	YES
Holiday FE	YES	YES	YES	YES	YES	YES	YES
% of avg	-0.04	-0.37	-0.06	0.25	-0.53	-0.06	0.38
$t(\text{HIGH}=\text{LOW})$		3.313			3.114		

Regression of daily crime rates on daily market returns by income and combined stockholder status from 1991 to 2015. High is defined as the top tercile of income (combined stockholder status) for all agency locations, while low is defined as the bottom tercile. Combined stockholder status is defined using the tercile sorted intersection of income, estimated stockholder propensity, and dividend income as a proportion of AGI. Market returns are divided by its trailing 252-day standard deviation. Crime rates are measured as the number of incidents per 100 million people. All incidents include all crime offenses in Table 1, while violent crime only include assaults and homicides. Controls are included in all specifications, but with coefficients reported in Table IA.4 of the Internet Appendix. Controls include cloud cover, temperature, humidity, precipitation, snowfall, snow depth, wind speed, ozone pollution, sunset hour, moon fraction, seasonal affective disorder, start and end of daylight savings time, geomagnetic storms, and current and previous day game outcomes for the NFL and MLB. The market return coefficient as a percentage of the dependent variables average is indicated in the penultimate row, while a subsample t -test for the equality of the coefficients for high- and low-income locations is provided in the last row. Parentheses contain t -statistics with heteroscedasticity robust standard errors clustered by time and agency. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 4: Returns: Daily lagging and leading market returns and violent crime rates

A. High-income

Returns lag/lead	$t-3$	$t-2$	$t-1$	t	$t+1$	$t+2$	$t+3$
Market return	1.226 (0.293)	6.118 (1.633)	-6.375 (-1.617)	-12.47*** (-3.291)	-3.028 (-0.819)	-0.139 (-0.0384)	3.842 (0.999)
Observations	1,679,140	1,676,679	1,690,712	1,707,967	1,689,527	1,671,057	1,674,893
R^2	.266	.266	.267	.258	.247	.255	.264
Controls	YES	YES	YES	YES	YES	YES	YES
Agency × Year FE	YES	YES	YES	YES	YES	YES	YES
Agency × Month of year FE	YES	YES	YES	YES	YES	YES	YES
Agency × Week of month FE	YES	YES	YES	YES	YES	YES	YES
Agency × Day of week FE	YES	YES	YES	YES	YES	YES	YES
Turn of month FE	YES	YES	YES	YES	YES	YES	YES
Holiday FE	YES	YES	YES	YES	YES	YES	YES
% of avg	0.04	0.18	-0.19	-0.37	-0.09	0.00	0.11
$t(\text{HIGH}=\text{LOW})$	0.601	1.995	0.682	3.313	1.023	0.554	0.200

B. Low-income

Returns lag/lead	$t-3$	$t-2$	$t-1$	t	$t+1$	$t+2$	$t+3$
Market return	6.385 (0.851)	-9.125 (-1.370)	-3.781 (-0.519)	15.22** (2.043)	5.001 (0.722)	4.675 (0.592)	2.102 (0.269)
Observations	1,145,570	1,144,101	1,150,652	1,154,058	1,137,676	1,130,968	1,139,926
R^2	.287	.287	.290	.279	.265	.278	.288
Controls	YES	YES	YES	YES	YES	YES	YES
Agency × Year FE	YES	YES	YES	YES	YES	YES	YES
Agency × Month of year FE	YES	YES	YES	YES	YES	YES	YES
Agency × Week of month FE	YES	YES	YES	YES	YES	YES	YES
Agency × Day of week FE	YES	YES	YES	YES	YES	YES	YES
Turn of month FE	YES	YES	YES	YES	YES	YES	YES
Holiday FE	YES	YES	YES	YES	YES	YES	YES
% of avg	0.10	-0.15	-0.06	0.25	0.08	0.08	0.03

Regression of daily violent crime rates on market returns from 1991 to 2015. Returns lag/lead indicates the timing of the market return relative to the violent crime rate at date t , where the date t crime rate includes nontrading days. Observations differ depending on availability of data for nontrading days. Market returns are divided by its trailing 252-day standard deviation. Violent crime rates are measured as the number of incidents per 100 million people. All specifications include the control variables listed in Table 3. The market return coefficient as a percentage of the dependent variables average is indicated in the penultimate (last) row in panel A (panel B), while a subsample t -test for the equality of the coefficients for high- and low-income locations is provided in the last row of panel A. Parentheses contain t -statistics with heteroscedasticity robust standard errors clustered by agency and time. * $p < .1$; ** $p < .05$; *** $p < .01$.

Read. The main table is the central result: market returns decrease violent crime in high-income locations and increase it in low-income locations. The lead-lag table shows that this is a contemporaneous same-day relationship rather than a delayed reaction or reverse-prediction pattern.

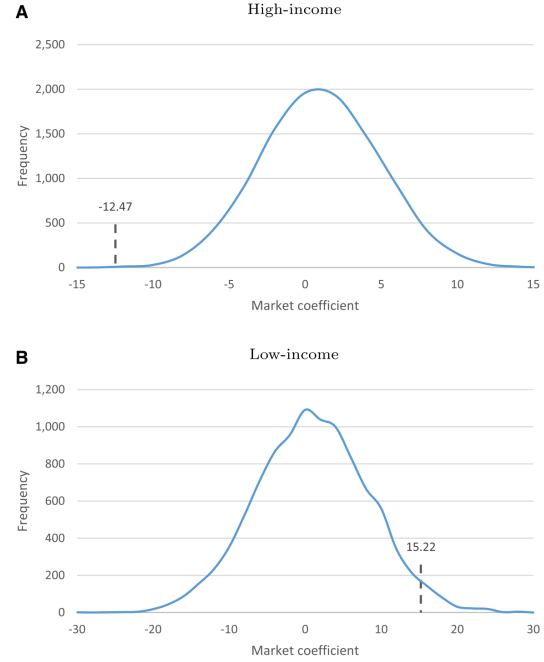
6.4 Table 5: Hourly timing of violent crime rates and Figure 3: Market returns and violent crime rates randomization test

Table 5: Hourly timing of violent crime rates

<i>A. High-income</i>						
Time (EST)	9AM – 4PM	4PM – 6AM	4PM – 7PM	7PM – 10PM	10PM – 2AM	2AM – 6AM
Market return	–2.713 (–1.364)	–9.469*** (–3.033)	–4.965*** (–3.113)	–1.955 (–1.286)	–1.757 (–0.988)	–0.793 (–0.747)
Observations	1,707,967	1,707,967	1,707,967	1,707,967	1,707,967	1,707,967
R^2	.121	.193	.079	.081	.099	.074
Controls	YES	YES	YES	YES	YES	YES
Agency × Year FE	YES	YES	YES	YES	YES	YES
Agency × Month of year FE	YES	YES	YES	YES	YES	YES
Agency × Week of month FE	YES	YES	YES	YES	YES	YES
Agency × Day of week FE	YES	YES	YES	YES	YES	YES
Turn of month FE	YES	YES	YES	YES	YES	YES
Holiday FE	YES	YES	YES	YES	YES	YES
% of avg	–0.28	–0.44	–0.86	–0.32	–0.25	–0.29
$t(\text{HIGH}=\text{LOW})$	0.771	5.020	2.379	2.719	2.784	2.301
<i>B. Low-income</i>						
Time (EST)	9AM – 4PM	4PM – 6AM	4PM – 7PM	7PM – 10PM	10PM – 2AM	2AM – 6AM
Market return	–5.949 (–1.609)	24.42*** (4.080)	3.310 (1.071)	7.393** (2.398)	8.930*** (2.625)	4.788** (2.196)
Observations	1,154,058	1,154,058	1,154,058	1,154,058	1,154,058	1,154,058
R^2	.118	.218	.082	.093	.112	.090
Controls	YES	YES	YES	YES	YES	YES
Agency × Year FE	YES	YES	YES	YES	YES	YES
Agency × Month of year FE	YES	YES	YES	YES	YES	YES
Agency × Week of month FE	YES	YES	YES	YES	YES	YES
Agency × Day of week FE	YES	YES	YES	YES	YES	YES
Turn of month FE	YES	YES	YES	YES	YES	YES
Holiday FE	YES	YES	YES	YES	YES	YES
% of avg	–0.37	0.60	0.32	0.65	0.66	0.87

Regression of violent crime rates by time of day on market returns from 1991 to 2015. The column header indicates the time in Eastern Standard Time (EST) that the violent crime occurred, while the market return is for the entire day and is divided by its trailing 252-day standard deviation. Violent crime rates are measured as the number of incidents per 100 million people. All specifications include the control variables listed in Table 3. The market return coefficient as a percentage of the dependent variables average is indicated in the penultimate (last) row in panel A (panel B), while a subsample t -test for the equality of the coefficients for high- and low-income locations is provided in the last row of panel A. Parentheses contain t -statistics with heteroscedasticity robust standard errors clustered by agency and time. * $p < .1$; ** $p < .05$; *** $p < .01$.

Figure 3: Market returns and violent crime rates randomization test



Read. The hourly timing table shows that the market-return relationship is concentrated after the market close. The randomization figure shows that randomly reassigned market returns do not reproduce the main coefficients, which supports the importance of actual same-day alignment.

6.5 Table 6: SUE and violent crime rates and Table 7: Market returns, alcohol use, and violent crime rates

Table 6: SUE and violent crime rates

<i>A. Surprise unexpected earnings</i>				
Income level	ALL	HIGH	MED	LOW
SUE	-1.370 (-0.906)	-4.475** (-2.319)	0.0132 (0.00478)	6.043* (1.662)
Observations	1,254,850	559,873	410,076	284,588
R ²	.250	.213	.249	.240
Controls	YES	YES	YES	YES
Agency FE	YES	YES	YES	YES
Date FE	YES	YES	YES	YES
% of avg	-0.03	-0.13	0.00	0.10
t(HIGH=LOW)		2.555		
<i>B. Announcement day returns</i>				
Income level	ALL	HIGH	MED	LOW
Announcement ret (0,1)	1.213 (1.191)	-1.976* (-1.686)	1.612 (0.912)	5.106** (2.032)
Observations	1,289,255	576,026	420,633	292,289
R ²	.249	.213	.249	.239
Controls	YES	YES	YES	YES
Agency FE	YES	YES	YES	YES
Date FE	YES	YES	YES	YES
% of avg	0.03	-0.06	0.03	0.08
t(HIGH=LOW)		2.554		

Regression of daily violent crime rates on daily standardized unexpected earnings (SUE) by income from 1995 to 2015. In panel A, SUE is defined as the difference between reported quarterly earnings and earnings from the same quarter in the previous year, scaled by the quarter end price. An equal weighted SUE for each region is calculated using salient firms with market caps greater than the NYSE 10th percentile of market capitalization, stocks with at least two analyst estimates in IBES, and stocks that have prices greater than \$1. Prior to equal weighting, SUE is winsorized at the 1st and 99th percentiles within each calendar quarter. Panel B uses the announcement day return defined as the two day cumulative return on announcement date t and $t+1$ for salient firms as defined previously. High-income is defined as the top tercile of income for all agency locations, while low-income is defined as the bottom tercile. Violent crime rates are measured as the number of incidents per 100 million people. All specifications include the location specific control variables listed in Table 3. The coefficient as a percentage of the dependent variables average is indicated in the penultimate row, while a subsample t -test for the equality of the coefficients for high- and low-income locations is provided in the last row. Parentheses contain t -statistics with heteroscedasticity robust standard errors clustered by police agency and date. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 7: Market returns, alcohol use, and violent crime rates

<i>A. Market returns and alcohol suspected violent crime</i>				
Income level	ALL	HIGH	MED	LOW
Market return	-0.312 (-0.266)	-0.574 (-0.449)	-0.956 (-0.472)	0.815 (0.325)
Observations	4,344,529	1,707,967	1,482,504	1,154,058
R ²	.113	.108	.113	.109
Controls	YES	YES	YES	YES
All FE	YES	YES	YES	YES
% of avg	-0.06	-0.14	-0.16	0.11
t(HIGH=LOW)		0.493		
<i>B. Alcohol and drug usage and violent crime</i>				
Income level	ALL	HIGH	MED	LOW
Market return	-1.566 (-0.435)	-12.53*** (-3.309)	-2.854 (-0.504)	15.73** (2.112)
Alcohol rate	0.0225*** (5.207)	0.0260*** (3.096)	0.0252*** (3.905)	0.0168** (2.220)
Drug rate	-0.0495*** (-16.73)	-0.0504*** (-15.99)	-0.0487*** (-13.37)	-0.0496*** (-7.542)
Observations	4,344,529	1,707,967	1,482,504	1,154,058
R ²	.294	.259	.284	.280
Controls	YES	YES	YES	YES
All FE	YES	YES	YES	YES
% of avg	-0.03	-0.37	-0.06	0.26
t(HIGH=LOW)		3.382		

Regression of violent crime rates on market returns from 1991 to 2015. Market returns are divided by its trailing 252-day standard deviation. Crime rates are measured as the number of incidents per 100 million people. Panel A regresses the daily alcohol suspected violent crime rate on market returns. Panel B regresses the violent crime rate on market returns, and local alcohol and drug usage. Local alcohol and drug rates are estimated using all alcohol and drug use suspected crime for Group A incidents, excluding violent crimes, but including the Group B crimes of driving under the influence and drunkenness. To prevent double counting, I have excluded crimes where both alcohol and drug use are suspected from the alcohol rate but included them in the drug rate. All specifications include the control variables listed in Table 3, as well as agency $\times$ year, agency $\times$ month of year, agency $\times$ week of month, agency $\times$ day of week, turn of month, and holiday fixed effects. The market return coefficient as a percentage of the dependent variables average is indicated in the penultimate row, while a subsample t -test for the equality of the coefficients for high- and low-income locations is provided in the last row. Parentheses contain t -statistics with heteroscedasticity robust standard errors clustered by agency and time. * $p < .1$; ** $p < .05$; *** $p < .01$.

Read. The earnings-surprise table uses local firm news with date fixed effects and finds the same high-versus-low income pattern. The alcohol table weakens a simple alternative explanation: the low-income result is not just mechanically driven by alcohol-suspected violent crime or by local alcohol/drug activity.

6.6 Table 8: Market returns and violent crime at home by known offender and Table 9: High-income violent crime rates by location, offender, and victim

Table 8: Market returns and violent crime at home by known offender

Income level	ALL	HIGH	MED	LOW
Market return	1.450 (0.640)	-6.193** (-2.335)	3.267 (0.904)	10.30** (2.005)
Observations	4,344,529	1,707,967	1,482,504	1,154,058
R^2	.179	.154	.173	.164
Controls	YES	YES	YES	YES
Agency $\times$ Year FE	YES	YES	YES	YES
Agency $\times$ Month of year FE	YES	YES	YES	YES
Agency $\times$ Week of month FE	YES	YES	YES	YES
Agency $\times$ Day of week FE	YES	YES	YES	YES
Turn of month FE	YES	YES	YES	YES
Holiday FE	YES	YES	YES	YES
% of avg	0.06	-0.37	0.13	0.33
t (HIGH=LOW)		2.853		

Regression of daily rate of violent crime committed at home by an offender that is known to the victim on market returns from 1991 to 2015. Market returns are divided by its trailing 252-day standard deviation. Violent crime rates are measured as the number of incidents per 100 million people. All specifications include the control variables listed in Table 3. The market return coefficient as a percentage of the dependent variables average is indicated in the penultimate row, while a subsample t -test for the equality of the coefficients for high- and low-income locations is provided in the last row. Parentheses contain t -statistics with heteroscedasticity robust standard errors clustered by agency and time. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 9: High-income violent crime rates by location, offender, and victim

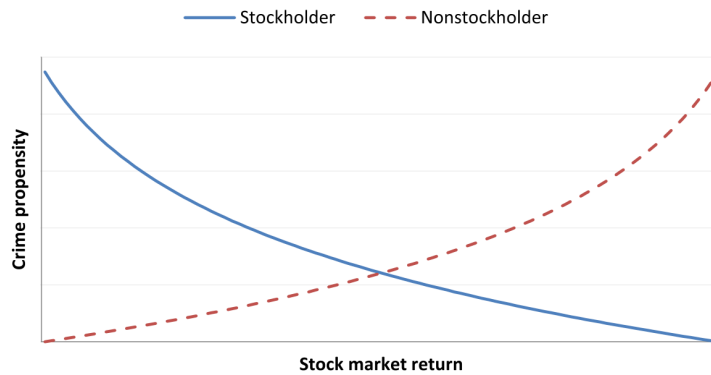
	Known	Known home	M Off known home	M Off F Vic known home	M Off F Vic intimate partner known home	Not M Off F Vic intimate partner known home
Market return	-8.850*** (-2.600)	-6.193** (-2.335)	-5.277** (-2.247)	-4.270** (-2.201)	-4.353*** (-2.670)	-8.119** (-2.400)
Observations	1,707,967	1,707,967	1,707,967	1,707,967	1,707,967	1,707,967
R^2	.231	.154	.127	.106	.081	.230
Controls	YES	YES	YES	YES	YES	YES
Agency $\times$ Year FE	YES	YES	YES	YES	YES	YES
Agency $\times$ Month of year FE	YES	YES	YES	YES	YES	YES
Agency $\times$ Week of month FE	YES	YES	YES	YES	YES	YES
Agency $\times$ Day of week FE	YES	YES	YES	YES	YES	YES
Turn of month FE	YES	YES	YES	YES	YES	YES
Holiday FE	YES	YES	YES	YES	YES	YES
% of avg	-0.34	-0.37	-0.42	-0.45	-0.70	-0.30
t (HIGH=LOW)	2.571	2.853	2.866	2.221	2.515	2.626

Regression of daily high-income violent crime rates by location, offender, and victim on market returns from 1991 to 2015. The column header indicates the dependent variable of the violent crime rate by location, offender, victim, and/or relationship between offender and victim. Violent crime rates are measured as the number of incidents per 100 million people. Market returns are divided by its trailing 252-day standard deviation. All specifications include the control variables listed in Table 3. The market return coefficient as a percentage of the dependent variables average is indicated in the penultimate row, while a subsample t -test for the equality of the coefficients for high- and low-income (from Table 1A.14) locations is provided in the last row. Parentheses contain t -statistics with heteroscedasticity robust standard errors clustered by agency and time. * $p < .1$; ** $p < .05$; *** $p < .01$.

Read. The home-known-offender table addresses the “more public targets” explanation. The offender-victim table shows that the high-income effect is especially large, in percentage terms, for male-on-female intimate-partner violence at home. This is consistent with stock-market losses creating household stress.

6.7 Figure 4: Stockholder and nonstockholder crime propensity

Figure 4: Stockholder and nonstockholder crime propensity



Read. The theoretical figure summarizes the relative-wealth mechanism. Positive market returns lower crime propensity for stockholders but raise it for nonstockholders because nonstockholders fall behind in relative wealth.

7 Short summary

- **Method.** Merge daily agency-level crime data with daily market returns and estimate high-frequency panel regressions with rich fixed effects and controls.
- **Main empirical result.** Positive market returns are associated with lower violent crime in high-income/investor locations and higher violent crime in low-income/noninvestor locations.
- **Timing result.** The effect is same-day and concentrated after the market close.
- **Identification support.** Local earnings surprises with date fixed effects produce the same pattern, helping rule out common daily shocks.
- **Mechanism.** The evidence is consistent with relative-wealth or social-comparison utility: investors gain from market increases; noninvestors can feel worse off when others gain.
- **Policy/social implication.** Stock-market fluctuations can impose externalities through psychological distress and crime, not only through financial wealth changes.

8 Conclusion

Core conclusion. Daily stock returns affect more than investor portfolios. They appear to affect psychological well-being and real behavior, as measured by violent crime.

Economic intuition. For investors, good stock-market news increases wealth and reduces stress, lowering violent crime. For noninvestors, good stock-market news can worsen relative status, raising stress and violent crime.

Takeaway. The stock market is socially salient even for people who do not own stocks. This means finance can generate psychological externalities: market gains and losses can reshape emotions, household conflict, and crime through both direct wealth and relative-status channels.